COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	Computer Science and Engineering & AIDS
Programme:	Computer Science and Engineering & AIDS
Course Code & Course Name	CSA1703 & ARTIFICIAL INTELLIGENCE
Academic Year / Batch	2026-2027
Faculty Name	DR. JEENA R
Assignment Title	Automated Sign Language Recognition and Speech Conversion System
Date of Issue	01/09/2026
Date of Submission	02/09/2026
Maximum Marks	100
Course Outcome(s) – CO	CO5
Bloom's Taxonomy Level	• **Evaluate (Level 5):** Comparing model architectures and trade-offs (latency vs. accuracy). **Create (Level 6):** Building an end-to-end vision, deep learning, and speech pipeline.
SDG Mapping (SDG 1-17)	**SDG 10 (Reduced Inequalities):** Removes communication barriers for deaf individuals. **SDG 9 (Innovation & Infrastructure):** Enables efficient edge AI on low-cost devices. **SDG 3 (Good Health):** Facilitates private doctor-patient communication without interpreters.
Industry / Societal Relevance	**Societal:** Restores personal autonomy and privacy in public, medical, and legal settings. **Industry:** Powers gesture-based HCI, spatial computing (AR/VR), and low-cost assistive kiosks.

B. Assignment Problem / Challenge

Communication between sign-language users and people who do not understand sign language can be difficult. Traditional communication methods may require a human interpreter or written communication.

The challenge is to develop an **automated sign language recognition system** that captures hand gestures using a camera, identifies the corresponding sign using an AI/ML model, converts the recognized sign into text, and finally produces speech output.

The system should provide accurate, fast, and user-friendly communication while considering variations in hand position, lighting, background, gesture orientation, and individual signing styles.

C. Problem Statement

Problem / Design Challenge

Design and develop an Automated Sign Language Recognition and Speech Conversion System using computer vision and machine learning techniques.

The system should:

1. Capture sign language gestures using a camera.
2. Detect and track the user's hand.
3. Extract relevant hand/gesture features.
4. Recognize the performed sign using a trained machine-learning/deep-learning model.
5. Convert the recognized sign into corresponding text.
6. Convert the text into speech using a text-to-speech system.
7. Provide the output in near real time.

Expected Output

Sign Gesture → Image/Video → Hand Detection → Feature Extraction → ML/DL Model → Recognized Text → Speech Output

D. Requirements and Constraints

Functional Requirements

- Camera-based gesture capturing.
- Hand detection and tracking.
- Sign-language gesture recognition.
- Text generation from recognized gestures.
- Text-to-speech conversion.
- Real-time or near-real-time processing.
- User-friendly interface.
- Recognition of multiple predefined signs.

Technical Requirements

- Python programming language.
- Computer vision library such as OpenCV.
- Machine-learning/deep-learning framework.
- Sign language dataset.
- Webcam/camera.
- Text-to-speech library/API.
- Sufficient computational resources for model training and inference.

Constraints

- Recognition accuracy may be affected by poor lighting.

- Complex gestures may be difficult to classify.
- Background objects may interfere with hand detection.
- Different users may perform the same gesture differently.
- Real-time processing requires sufficient computational performance.
- The system may initially support only a predefined vocabulary.

The assignment format specifically recommends considering functional, performance, technical, cost/time, reliability, sustainability, ethical, user, power, security/privacy, and other relevant constraints.

E. Student Work

1. Problem Understanding and Formulation

The primary problem is the communication barrier between sign-language users and non-sign-language users.

The proposed system attempts to bridge this gap through an automated AI-based solution.

Input

- Live video from webcam/camera
- Hand and finger movements
- Sign-language gestures

Processing

- Image acquisition
- Hand detection
- Image preprocessing
- Feature extraction
- Gesture classification
- Text generation

Output

- Recognized sign as text
- Corresponding speech/audio output

Basic System Flow

User Performs Sign

↓

Webcam

↓

Image / Video Capture

↓
Image Preprocessing
↓
Hand Detection
↓
Feature Extraction
↓
ML / Deep Learning Model
↓
Sign Recognition
↓
Text Output
↓
Text-to-Speech
↓
Audio Output

2. Application of Course Knowledge

The system applies concepts from:

Computer Vision

Computer vision techniques are used to capture and process hand gestures from images or video.

Machine Learning

A classification model can be trained to identify different sign-language gestures.

Possible algorithms include:

- Support Vector Machine (SVM)
- Random Forest
- K-Nearest Neighbors (KNN)
- Convolutional Neural Network (CNN)

Deep Learning

A CNN can automatically learn important visual features from sign-language images and classify them into different gesture classes.

Natural Language Processing

Recognized signs can be converted into meaningful text or sentences.

Speech Processing

A Text-to-Speech (TTS) component converts the generated text into audible speech.

3. Solution / Design / Methodology

Step 1: Data Collection

A sign-language dataset containing different hand gestures is collected.

Example classes:

A

B
C
D
E
...
Z
Additional classes can include:
Hello
Thank You
Yes
No
Help
Good Morning
I Love You

Step 2: Image Acquisition
A webcam captures the user's hand gesture continuously.
Camera
↓
Live Video Stream
↓
Individual Frames

Step 3: Preprocessing
The captured images are processed to improve recognition.
Operations may include:

- Resizing
- Noise reduction
- Normalization
- Background handling
- Image enhancement

Step 4: Hand Detection
A hand-detection algorithm identifies the hand region from the captured frame.
The detected hand is then used as the input for gesture recognition.

Step 5: Feature Extraction
Important features are extracted from the hand gesture, such as:

- Finger positions
- Finger angles
- Hand orientation
- Hand shape

- Landmark coordinates

These features help the model distinguish between different signs.

Step 6: Model Training

The extracted data is divided into training and testing datasets.

Sign Language Dataset
↓
Preprocessing
↓
Feature Extraction
↓
Training Data ──→ ML/DL Model
↓
Testing Data
↓
Model Evaluation

A CNN-based classifier can be used for image-based sign recognition.

Step 7: Sign Recognition

When a user performs a gesture in front of the camera, the trained model predicts the corresponding sign.

For example:

Hand Gesture
↓
AI Model
↓
"HELLO"

Step 8: Text Conversion

The recognized signs are displayed as text.

Example:

Sign 1 → HELLO
Sign 2 → HOW
Sign 3 → ARE
Sign 4 → YOU

Output:
"HELLO HOW ARE YOU"

Step 9: Speech Conversion

The generated text is passed to a Text-to-Speech system.

Recognized Text

↓
Text-to-Speech Engine
↓
Audio Signal
↓
Speaker
The user can therefore communicate through spoken audio without requiring a human interpreter.

4. Use of Modern Tools
The following tools can be used:

Tool	**Purpose**
Python	Main programming language
OpenCV	Image/video processing
MediaPipe	Hand detection and landmark tracking
TensorFlow/Keras	Deep-learning model development
NumPy	Numerical processing
Pandas	Dataset processing
Matplotlib	Graphs and visualization
Scikit-learn	Machine-learning algorithms
Text-to-Speech	Speech generation
Jupyter Notebook / VS Code	Development environment

The assignment format identifies Python, TensorFlow and other modern computing tools as suitable examples for CSE/IT assignments.

5. Results and Validation
The performance of the system can be evaluated using:
Accuracy
Measures the percentage of correctly recognized signs.
Precision
Measures how accurately the model identifies a particular sign.
Recall
Measures how many actual signs are correctly recognized.
F1-Score
Provides a combined measure of precision and recall.
Confusion Matrix
A confusion matrix can be used to identify which signs are correctly or incorrectly classified.
Example Result Table

Sign	Actual	Predicted	Result
A	A	A	Correct
B	B	B	Correct
C	C	C	Correct
D	D	B	Incorrect
E	E	E	Correct

The assignment format recommends presenting results through tables, graphs, screenshots, performance metrics, and test results, followed by validation against the stated requirements.

6. Analysis and Engineering Decision

A traditional sign-language recognition approach may depend heavily on manually designed features. A deep-learning-based approach can automatically learn useful visual features from training data.

Possible Comparison

Method	Advantages	Limitations
Manual Feature Extraction	Simple and interpretable	Less flexible
SVM	Good for smaller datasets	Requires feature engineering
Random Forest	Easy to implement	May not handle complex images well
CNN	Strong image-recognition capability	Requires more training data/computation

Selected Approach

A **CNN-based sign recognition model combined with computer vision and text-to-speech conversion** can be selected because the system deals primarily with visual gesture information. The final choice should be justified using measured accuracy, processing speed, resource requirements, and recognition performance, as required by the assignment structure.

7. Broader Considerations

Social Impact

The system can reduce communication barriers and improve interaction between sign-language users and the wider community.

Accessibility

It can support communication in:

- Schools
- Colleges
- Hospitals
- Government offices
- Banks
- Public-service centers

- Workplaces

Ethical Considerations

- User video data should be handled responsibly.
- Personal information should not be unnecessarily collected.
- Dataset diversity should be considered.
- The model should be tested with different users and signing styles.
- Recognition errors should not be treated as guaranteed interpretations.

Privacy

If the system processes live camera input, video should preferably be processed locally when possible, and unnecessary recording or storage should be avoided.

Sustainability

A lightweight model can reduce computational requirements and enable deployment on affordable systems.

The assignment format explicitly includes society, safety, ethics, economics, accessibility, sustainability, and professional responsibility as relevant broader considerations.

8. Conclusion

The **Automated Sign Language Recognition and Speech Conversion System** provides an AI-based solution for reducing communication barriers between sign-language users and non-sign-language users. The proposed system combines computer vision, hand detection, machine learning/deep learning, text generation, and text-to-speech technologies.

The system captures hand gestures through a camera, identifies the performed sign using an AI model, converts it into text, and produces corresponding speech output. Although challenges such as lighting conditions, background variation, gesture similarity, and differences in individual signing styles can affect recognition performance, the system can be improved through larger and more diverse datasets, better model architectures, and real-time optimization.

9. Student Reflection

Through this assignment, I learned how artificial intelligence and computer vision can be applied to solve real-world accessibility problems. I gained knowledge about image processing, hand-landmark detection, machine-learning model training, classification, and text-to-speech conversion.

I also understood that developing an AI system requires more than simply training a model. Dataset quality, preprocessing, model evaluation, user requirements, privacy, accessibility, and real-world testing are equally important.

If additional time and resources were available, the system could be improved by supporting a larger sign-language vocabulary, continuous sentence recognition, multiple regional sign languages, facial expressions, two-hand gestures, and deployment on mobile or edge devices.

10. References

1. Research papers and publications related to automated sign-language recognition.
2. Documentation for OpenCV and computer vision techniques.

3. Documentation for MediaPipe hand landmark detection.
4. TensorFlow/Keras documentation for deep-learning model development.
5. Scikit-learn documentation for machine-learning algorithms.
6. Text-to-Speech technology documentation.
7. Sign-language image/video datasets used for model training and evaluation.
8. Any AI-assisted tools, datasets, software libraries, and external resources used during implementation should be properly acknowledged, as specified in the assignment format.

Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	**100**

F. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO5	PO1, PO2	L3/L4	10
Application of knowledge	CO5	PO1, PO3	L3/L4	20
Solution / Design / Implementation	CO5	PO3, PO5	L4/L5/L6	20
Modern tool usage	CO5	PO5	L3/L4	10

Validation	CO5	PO4, PO5	L4/L5	15
Analysis & justification	CO5	PO2, PO4	L4/L5	15
Broader considerations	CO5	PO6, PO7	L4/L5	5
Documentation & reflection	CO5	PO9, P10	L3/L4	5

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is **not required to map to every PO** .

G. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	≥ 70%
Level 2	60–69%
Level 1	50–59%
Level 0	< 50%

Target CO Attainment: >70%

Actual CO Attainment: 89%

Faculty Analysis

Areas in which students performed well: Students demonstrated good technical skills in completing the assignment.

Areas requiring improvement: Some students required improvement in implementation details.

Corrective / Improvement Action: Provided individual feedback and technical support.

Action to be implemented in the next offering: Conduct short demonstrations before the assignment.

Documents to be Maintained

The department/course faculty shall maintain:

1. Assignment question/problem statement
2. CO–PO and Bloom's mapping
3. Assessment rubric
4. Student submissions
5. Tool/simulation/experimental evidence, where applicable
6. Evaluated scripts/reports
7. Marks and rubric-wise assessment
8. CO attainment analysis
9. Sample student work representing different performance levels
10. Faculty analysis and continuous-improvement action